

Muhammad Ahmad

AI Engineer

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SUMMARY

AI engineer with hands-on production experience shipping autonomous agent systems and real-time voice AI end to end. Built and deployed 7+ production systems in the past year, including a voice agent reaching sub-1-second latency. Strong backend foundation (FastAPI, multi-database systems) paired with rigorous system design and production hardening practices.

EXPERIENCE

VaconAI

Sept 2025 – Aug 2026

AI Engineer, Full-time

Remote

- Designed and shipped 7+ production AI/ML systems end to end across a multi-client engineering practice, owning architecture, implementation, and deployment from spec to production.
- Built a knowledge-graph service spanning three databases (**Neo4j**, **MongoDB**, **Pinecone**; 6 node types, 15 relationship types) that converts legal deposition transcripts into queryable contradiction and admission graphs in under 2 minutes end to end.
- Designed a two-stage extraction pipeline for the graph: deterministic parsing generates 200–300 nodes per transcript at zero LLM cost, then a concurrency-capped LLM pass adds 70–150 relationship nodes.
- Automated dispute resolution for an e-commerce payments platform with a 10-agent LLM routing system and a 4-tier offer-escalation ladder, replacing manual case handling with a bounded, trackable negotiation flow.
- Built automatic evidence pre-collection with a 3-tier order-lookup fallback (custom field → invoice ID → storefront search), staging dispute-specific evidence before any agent call runs.
- Built a real-time “co-counsel” feature streaming live transcription over **WebSocket** with a lease-based failover worker pool, and a 6-language automated return-processing pipeline for a customer-support platform.
- Built a production real-time voice agent with swappable local/cloud STT and TTS backends, reaching 0.7–1.0s voice-to-voice latency (vs. 2.4s all-local); extended to live phone calls via **Twilio** and a browser softphone.
- Built a six-phase autonomous content pipeline (keyword research, pillar/cluster mapping, content briefs, generation, on-page optimization) via **Claude Code CLI** in headless mode, then evolved it into a 24/7 self-operating system across 13 sites and 16 industries with a human-approval gate; published 364+ verified-live articles, growing organic search impressions 24% week over week.
- Built an **MCP** server exposing 17 tools across 9 research sources (Hacker News, Reddit, X, YouTube) with an offline-first cache of 8,400+ indexed transcripts.
- Engineered a resilient transcript-ingestion pipeline across 20 **YouTube** channels (1,000 candidates → 457 corpus chunks) with adaptive backoff.

Techomnis

2024 – Sept 2025

Software Developer, Full-time

Remote

- Built a medical transport management system processing 500+ daily trips, with a reporting dashboard, data exports, and a CSV bulk-import pipeline with validation.
- Implemented backend APIs with advanced filtering, pagination, and aggregation for client management; delivered a private production deployment with invite-only access control.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, C++

AI/ML & Agents: OpenAI Agents SDK, LLM Orchestration, RAG, MCP, PyTorch, Scikit-learn, LoRA Fine-Tuning, Cross-Encoder/SetFit Distillation, NLP

Backend & Infra: FastAPI, Flask, Docker, Modal, AWS, Google Cloud Platform, Async/Concurrent Processing

Data & Storage: Neo4j, MongoDB, Pinecone, PostgreSQL, SQLite FTS5, Apache Spark

Real-Time & Voice: Pipecat, LiveKit, WebRTC, Twilio, Deepgram, Whisper

Tools: Git, Linux, CI/CD, Streamlit

EDUCATION

Punjab University College of Information Technology (PUCIT)

2020 – 2024

B.S. Computer Science

Lahore, Pakistan

- Key coursework: Machine Learning, Data Structures, Database Systems, Statistics, Computer Vision.